서송병원 Forecasting Result Report

LAVAR Seed Stability, Blend Robustness, and Ensemble Interval Results

LAVAR Health-Economic Forecasting Study

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Abstract

This report summarizes the 서송병원 forecasting results produced by the Seosong replication workflow. The evaluation uses daily patient features and prescription targets from 2020-12-14 through 2025-12-31, with 1,844 daily observations, 38 patient-side covariates, and 3,126 prescription targets. Across five random seeds and 79 rolling folds per seed, the S11 latent-16 configuration improves average MSE over the S04 GRU configuration, but with materially higher seed-to-seed variance and tail risk. A fixed ensemble, $0.55\times$S11 $+0.45\times$S04, further reduces mean MSE relative to pure S11 by 0.0596, or 3.17%, but does not satisfy the promotion rule because p95 fold MSE worsens relative to pure S11.

1 Data and Evaluation Setup

The source artifacts are the Seosong report outputs from the seed stability, blend robustness, and ensemble workflows. The base data files are:

- `data_patient.parquet`: 1,844 rows by 38 columns.
- `data_prescription.parquet`: 1,844 rows by 3,126 columns.
- Date range: 2020-12-14 to 2025-12-31.

Each scenario was evaluated over five seeds: $\{42, 123, 456, 789, 1024\}$. The rolling evaluation used a 14-day forecast horizon, 14-day fold step, 90-day retraining cadence, quality triggers enabled, and 79 folds per run.

2 Model Scenarios

The Seosong workflow compares two base scenarios and one fixed blend:

- **S04_GRU_H32_L1**: GRU temporal decoder baseline.
- **S11_LATENT16**: latent dimension 16 configuration.
- **Blend w=0.55**: $0.55\times$S11 $+0.45\times$S04, clipped to non-negative predictions.

The ensemble interval artifact computes empirical 95% intervals using the five seeded blend predictions.

3 Seed Stability Results

Table 1 reports seed-level stability summaries from `seed_stability_summary.csv`. S11 has the better mean MSE and MAE, but its coefficient of variation for mean MSE is roughly six times larger than S04's, indicating a less stable seed response.

Table 1: Seed stability by base scenario.

Scenario	Seeds	Mean MSE	SD MSE	CV MSE	Mean MAE	p95 Fold MSE
S04 GRU H32 L1	5	2.0234	0.0540	0.0267	0.2712	4.6293
S11 LATENT16	5	1.8810	0.2986	0.1587	0.2616	4.4179

The S11 mean MSE improvement over S04 is 0.1424, equal to a 7.04% reduction relative to S04. However, S11's maximum fold MSE reaches 20.8928 in its worst seed, compared with 13.7482 for S04.

4 Blend and Ensemble Results

Table 2 summarizes the fixed blend against pure S04 and pure S11 from the ensemble metrics table. The blend is the best option by average MSE and average maximum-fold MSE, but pure S11 remains best by p95 fold MSE.

Table 2: Overall ensemble comparison.

Label	Seeds	Mean MSE	SD MSE	Mean MAE	p95 Fold MSE
Blend w=0.55	5	1.8214	0.1641	0.2615	4.4676
Pure S04	5	2.0210	0.0520	0.2708	4.6053
Pure S11	5	1.8810	0.2986	0.2616	4.4179

Relative to pure S11, the fixed blend reduces mean MSE by 0.0596 (-3.17%) and reduces the seed-level standard deviation of mean MSE from 0.2986 to 0.1641. Relative to pure S04, the blend reduces mean MSE by 0.1995 (-9.87%). The tradeoff is tail performance: blend p95 fold MSE is 4.4676, which is 0.0497 higher than pure S11.

5 Robustness Decision

The fixed-weight robustness decision table reports that both $w = 0.50$ and $w = 0.55$ win on mean MSE for three of five seeds, win on p95 fold MSE for two of five seeds, and win both criteria for only two of five seeds. Accordingly, `promote_blend_rule=False` for both weights.

Table 3: Blend promotion decision.

w	Seeds	Mean-MSE Wins	p95 Wins	Both Wins	Mean ΔMSE
0.50	5	3	2	2	-0.0538
0.55	5	3	2	2	-0.0596

This means the blend is useful as an average-error reducer and variance stabilizer, but it should not replace pure S11 as the default operating rule if tail-fold robustness is the governing criterion.

6 Interval Output

The final ensemble workflow generates per-drug empirical 95% intervals from seeded blend predictions: `ensemble_ci_per_drug.csv`. The visualization in Figure 1 plots true values, ensemble predictions, and the empirical interval for selected drug targets.

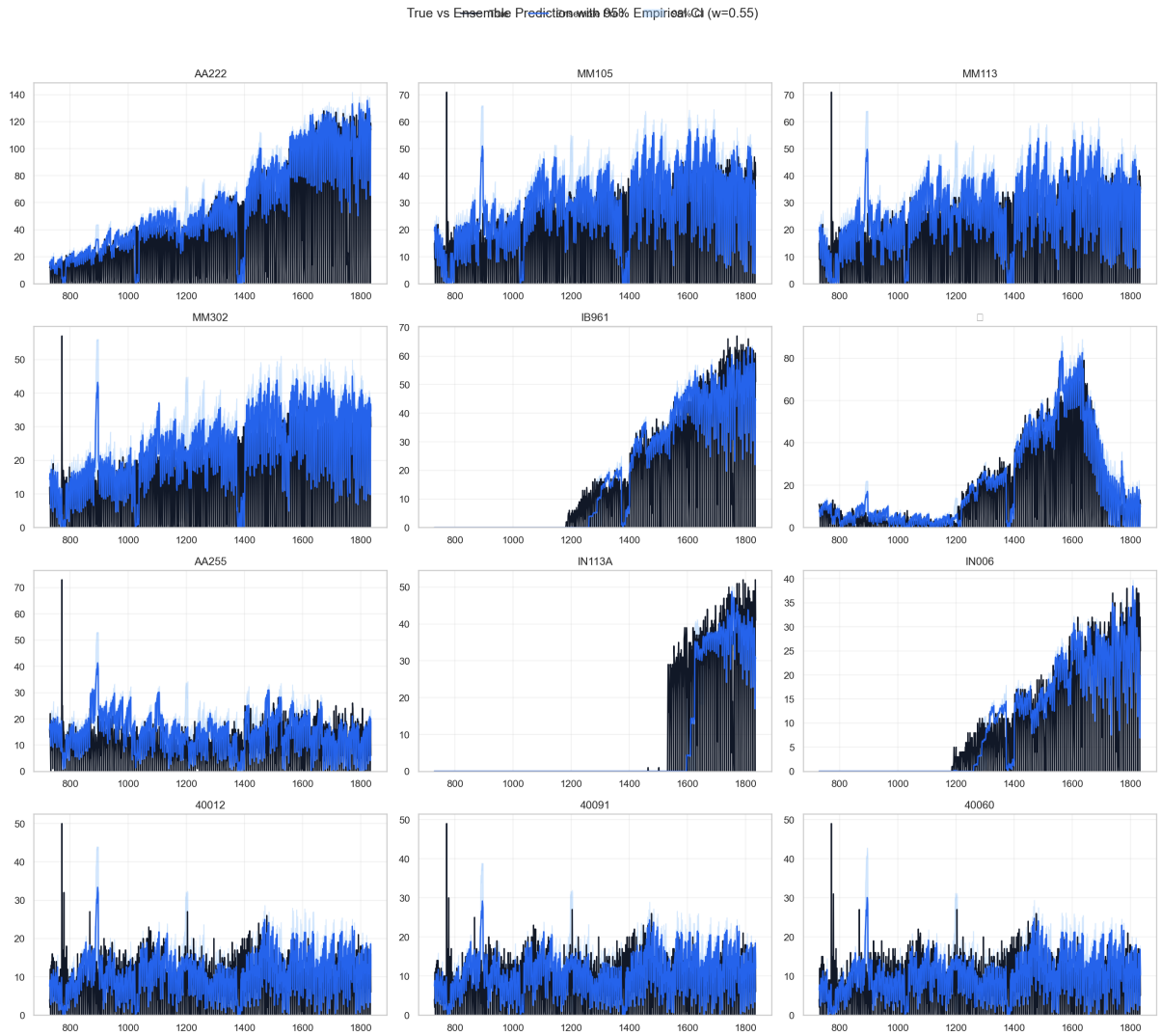


Figure 1: Seosong ensemble true vs. predicted values with empirical 95% intervals.

7 Conclusion

For 서송병원, S11 LATENT16 is the stronger single model by average error, while S04 is more seed-stable. The $w = 0.55$ blend gives the best average MSE and substantially reduces S11 seed variance, but it worsens p95 fold MSE against pure S11 and therefore fails the current promotion rule. The recommended reporting stance is:

- use S11 as the primary single-model benchmark;
- retain the $w = 0.55$ blend as a variance-stabilized sensitivity analysis and interval-production model;
- avoid promoting the blend as the default model until tail-fold robustness improves.